Ancient Chinese Timber Architecture. I: Experimental Study

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Abstract: As an important part of the civilization in China, ancient Chinese architecture has an indigenous and unique system of construction that has retained its principal characteristics from prehistoric times to the present and spread its influence to other countries such as Korea and Japan. With the exacerbating environmental encroachment and time impacts that have been reducing their structural resistance, it has become imperative to preserve this architectural heritage. To provide the basic characteristics of the timber structures as fundamental information for research and preservation, on-site measurements and model tests were conducted on the front tower over the North Gate of the Xi'an City Wall. In the full-scale on-site tests, the vibration modes of the tower were measured. The first four natural frequencies of the structure were found to be 1.10, 1.70, 2.73, and 3.10 Hz, respectively. Multipoint excitation tests on wooden and polymethyl methacrylate (Perspex) models validate that the first and second modes obtained by the on-site full-scale test are the vibration modes of the tower. Many other valuable data and information were obtained through the tests, such as the damping ratio and the hysteresis curves.

INTRODUCTION

Ancient timber architecture is an important part of the Chinese civilization, which has profound influence on some Asian countries such as Korea and Japan. Many of them are assigned as historical heritage sites, such as Fuogong Temple in Shanxi, China (built in 1056), Horioji Temple in Nala, Japan (built in 682), and Nam Dae Mun in Seoul, South Korea (rebuilt in 1448).

Unlike other kinds of structures, ancient Chinese timber architecture (Fig. 1) has the following unique characteristics (Chinese Ancient Architecture 1985):

- Dou gong (corbel bracket) (Fig. 2) is a bracket system in which each bracket is formed by a double bow-shaped arm, called gong, which supports a block of wood, called dou, in each side. This system is inserted between the top of a column and a cross beam as connections (Fig. 3). It functions not only as a special structural part to support the roof but also as the typical means for decoration.
- Members, such as beams and columns, are connected by tenon joints (plug-slot type connection), with the tenon as one component and mortise as another (Fig. 4). Tenon joints have various types, dimensions, and shapes for different functions.
- Columns are simply supported by natural cornerstones (Fig. 5).
- No nails or bracing are used in the timber frame.

As time passes, research and preservation of these kinds of historical structures become more and more necessary and important. To take effective measures to protect them, clear understanding of their current situation and structural properties is essential. Traces of the earliest Chinese timber architecture is found to be 5,000 years old, and the first book on the construction of timber structures (Li 1100) was published in the Song Dynasty. Liang (1984) had worked on ancient timber structures for many years in the early part of the last century, mainly focusing on their architectural and structural styles. Ma (1991) summarized their construction methods and structural characteristics. Wang's research (1992) focused on the distribution and transmission of the vertical loads in the timber structural members. King et al. (1996) studied the static linear and nonlinear properties of one type of plug-slot type connections by both experimental and numerical modeling methods. They pointed out that the semirigid connection's stiffnesses were seriously decreased by a component's surface deterioration. Because various kinds of plug-slot type connections were used in Chinese timber structures during thousands of years of construction, it is believed that no accurate value can be estimated to describe the stiffness of the connections.

Hayashi et al. (1998) presented their research work on the restoring properties of a wooden frame with masu-gumi (dou gong) and plug-slot connections (tenon joint) in an ancient Japanese timber structure. They found that embedding of the edges of the masu (bearing block) and the slip between each member cause most of the structural deformation. They also found that the structural performance varies with the level of vertical load. Yanagisawa et al. (1998) have conducted cyclic load and vibration tests on a true Japanese temple. The natural frequency and damping factor of the first mode were observed by constant microvibration, vibrator-forced vibration, and manpower-forced vibration tests. The natural frequencies were found to be within 1–2 Hz and the damping factor to be within

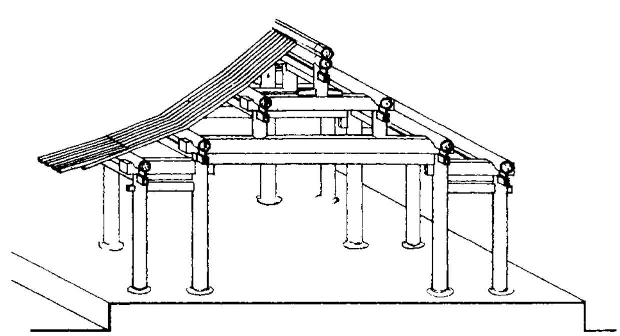

FIG. 1. Typical Ancient Timber Architecture

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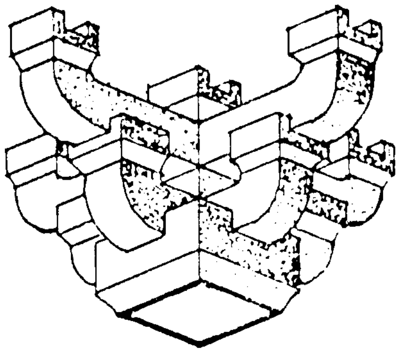

FIG. 2. Dou Gong (Corbel Bracket)

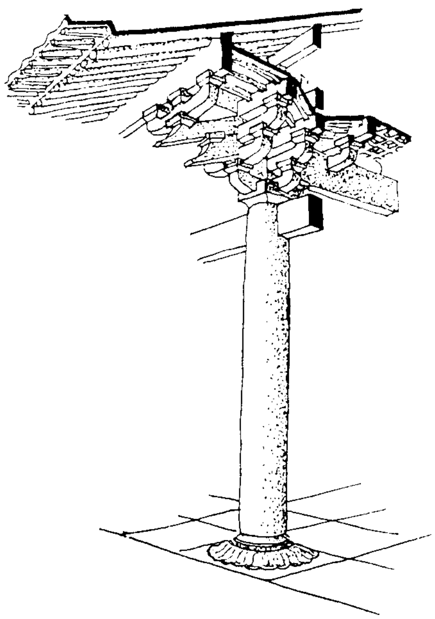

FIG. 3. Corbel Bracket Is Used as Connection of Roof and Columns

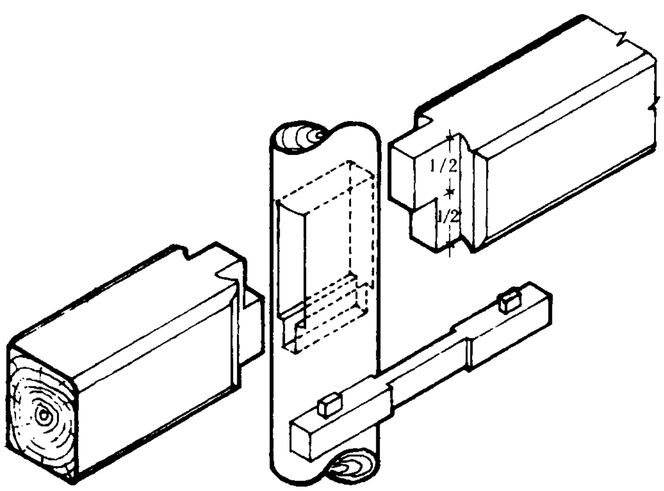

FIG. 4. Tenon Joint Is Used to Connect Beams and Columns

1–3%. Maekawa et al. (1998) conducted microtremor measurement on six one-floor traditional Japanese wooden houses and found that the average primary natural frequency was 3 Hz and the damping factor was 1–2%. Uchida et al. (1998) conducted more measurements on one- and two-story traditional timber buildings and multistory pagodas in Japan. The natural frequencies were found within 1.45–2.15 Hz for one- or two-story buildings and within 0.85–0.9 Hz for pagodas. A horizontal lumped mass system model was developed applicable to one- or two-story buildings, and a vertical lumped mass system model was developed applicable to pagodas. Seo et al. (1999) conducted experimental research on the static and cyclic behavior of Korean ancient wooden frames with tenon joints under lateral load. Although the structures in their work have similar characteristics to the ancient Chinese timber architectures, no dou gong (corbel brackets) were used.

With regard to research on timber structures, attention has been focused on the connections that made timber structures different. Brungraber (1985) tested joints and a few frames. He performed 2D finite-element analyses on joint details and proposed a three-spring joint model for frame analysis. The Subcommittee on Wood Research of the ASCE Committee on Wood (1986) highlighted that joints often are the weakest link in wood structures. A joint transfers shear, compression, and tension loads from one structural element to another. A large variety of joint-type configurations is one of the reasons for the limited research results on the stiffness, especially of the ancient timber joints. During the past decade, modern analytical procedures such as the finite-element method have been applied to predict deflections and the ultimate strength of components. Testing on full-size components and structures, which has often been conducted because of unavailable analytical procedures, usually provides specific solutions with no general application. However, there is great demand to develop a rigorous analytical model to accurately predict structural behavior. Chui et al. (1998) suggested a finite-element model for nailed wood joints under reversed cyclic load. Bulleit et al.

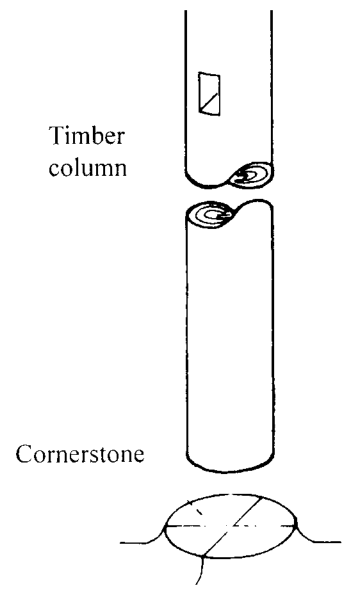

FIG. 5. Column Simply Supported on Cornerstone

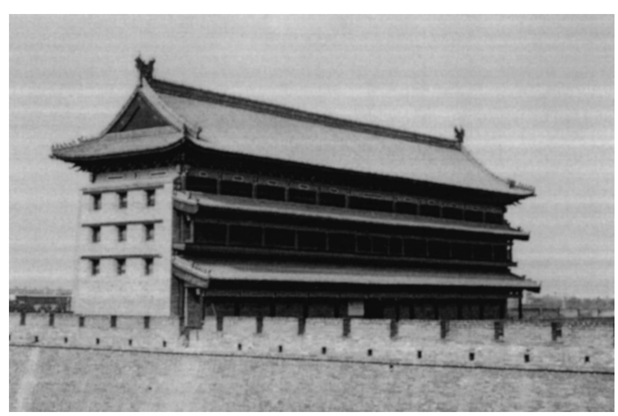

FIG. 6. Front Tower over North Gate of Xi'an City Wall (Tower)

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(1999) presented their research on the behavior and modeling of wood-pegged timber frames in response to the need driven by more renovation and rehabilitation of historic wood structures. Joint and frame models based on thorough examination of various types of tenon joints were suggested. Recently, Sandberg et al. (2000) presented more experimental data and analytical models to assist in the analysis and design of pegged joints in traditional timber frames in North America.

Research has been conducted on the dynamic characteristics of ancient Chinese timber architecture by Fang (1993). The Front Tower over the North Gate of the Xi'an City Wall (referred as the "Tower") (Fig. 6), a typical historic ancient timber structure located in Xi'an City, China, was selected as a research case. To enhance the understanding of dynamic and seismic properties of ancient timber structures and to provide information and suggestions for their protection and maintenance, on-site measurement and model tests have been conducted.

ON-SITE FULL-SCALE TEST

The Tower was built from 1370 to 1378 during the Ming Dynasty. It was constructed on top of the North Gate of the 12-m-high Xi'an City Wall, with the ridge being 18.14 m in height (Fig. 7), 50.6 m long, and 10.52 m wide. Its timber structure is supported by 36 wooden columns (12 $\times$ 3), which are 0.55 m in diameter (Fig. 8). Structural members are connected by the dou gong (corbel brackets) and tenon joints. It endured the earthquakes in 1556 and 1654 without serious damage. During the on-site measurement and investigation, the building was under maintenance construction. Some wooden members have decayed because of poor drainage and ventilation. Standard wood material tests (Cheng et al. 1985) on the specimens from uncorrupted wooden members (600-year-old *pinus armandi*) and wooden members (*P. Koraiensis*) used for maintenance were carried out. Table 1 shows the results. Obviously, the uncorrupted wooden members used to build the structure 600 years ago indicate no great difference in quality from the wooden members used today.

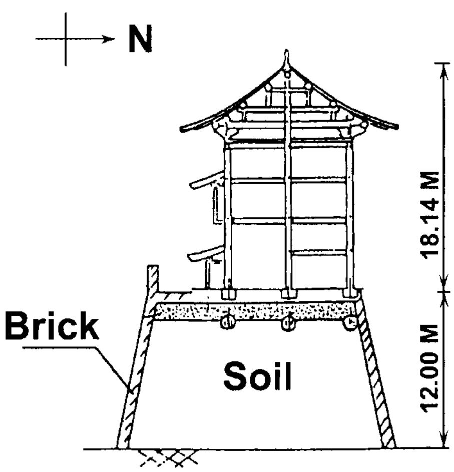

FIG. 7. Elevation of Tower

In the experiment, the environmental influence was chosen as the input signal and the structural microtremor was recorded as the response signal. The method is suitable to the situation in which excitation is not possible or is not allowed. Acceleration sensors were installed on columns 2, 4, 6, 7, 9, and 11 at points I, II, III, IV, V, and VI with the heights of 1.5, 4.0, 7.0, 10.0, 14.5, and 16.0 m, respectively (as shown in Fig. 9). Because the structure is symmetrical, the basic vibration modes, the axial vibration mode along Y (symmetrical) and the torsion mode about Z (antisymmetrical), are studied.

The following spectra and functions, which can be obtained by processing the signals picked up through an acceleration sensor by fast Fourier transform analyzer, are examined (Tanikuchi 1991):

- Fourier spectra—The resonant point can be found at the frequency where peaks could be observed.
- Cross-correlation function—Its phase lag-frequency curve can indicate the relative vibration direction of two measured points. If the phase lag at a certain frequency is close to 0°, the two points vibrate in the same direction. If the phase lag is close to 180°, the two points vibrate in opposite directions.
- Coherence function—The level of correlation of signals from two points is indicated. When its value reaches 1.0, the signals at the two points are coherent; i.e., the signals are the undisturbed output of the same excitation.

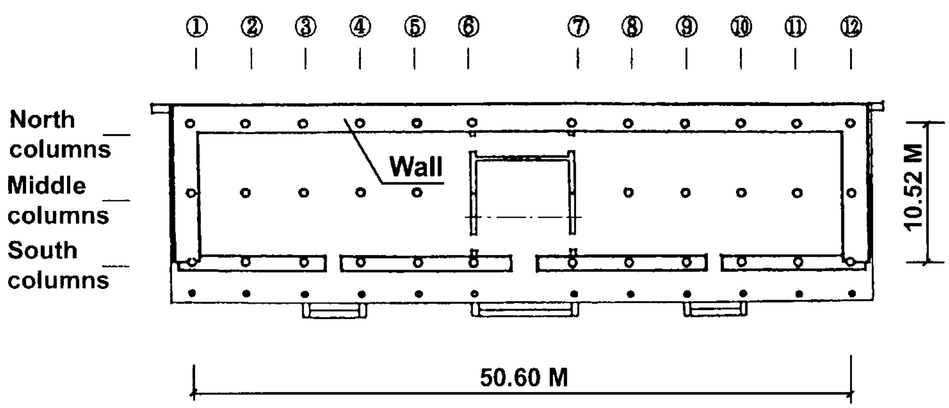

FIG. 8. Plan View of Tower

TABLE 1. Comparison of Properties between Original Uncorrupted Wood and New Wood Used in Maintenance

Timber	γ_{15} (kg/m^3)	E (kN/m^2)	σ_c (kN/m^2)	σ_l (kN/m^2)	σ_b (kN/m^2)	τ_b (kN/m^2)
Original: *Pinus armandi*	410	83,035.4	48,294.4	84,299.6	41,797.0	8,192.8
New: *P. koraiensis*	450	87,906.0	34,995.8	100,597.0	68,796.0	8,300.6

Note: γ_{15} = density of wood with 15% moisture content; E = Young's modulus of elasticity; σ_c = compression strength; σ_l = tensile strength; σ_b = bending strength; and τ_b = shear strength.

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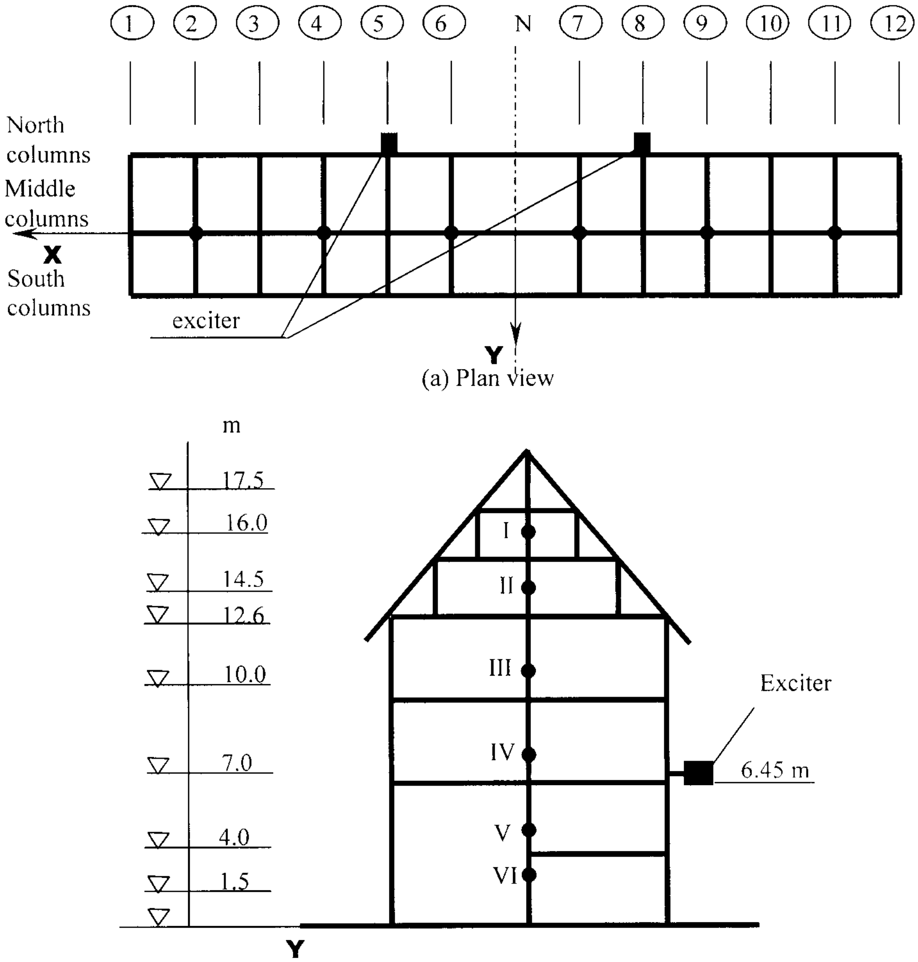

FIG. 9. Positions of Sensors and Exciters

Two on-site measurements were carried out. The first test was conducted during the maintenance construction, when the mud and tiles of the roof were removed and the roof was 80 ton lighter than normal. The signals of point III on column 2 (A) and point III on column 11 (B) were recorded and analyzed. The Fourier spectrum (P-f), phase-lag–frequency curve (TP-f), and coherent function curve (Co-f) are shown in Fig. 10. The frequency corresponding to the first peak in the Fourier spectrum of the signals from points A and B is observed as 1.45 Hz. The phase-lag–frequency curve (TP-f) is 13.9° and close to 0°; therefore, the directions of the two points are regarded as the same. The coherent function is 0.9995, so the two signals are coherent. To verify that 1.45 Hz is one of the natural frequencies of the structure, the above procedure was repeated at other symmetrical points (such as point III in column 4 and point III in column 9), and the same conclusion was reached. Therefore, 1.45 Hz is proved to be the first natural frequency of the structure with its roof 80 ton lighter than its normal condition, and its correspondent vibration mode is symmetrical (to the Y-axis).

The frequency corresponding to the second peak in the Fourier spectrum of the signals from points A and B is observed as 2.18 Hz. The phase lag of coherent density-function–frequency curve (TP-f) is $-165°$ and close to $-180°$; therefore, the directions of the two points are opposite. The coherent function is 0.999, so the signals are the undisturbed output of the same excitation. To confirm that 2.175 Hz is one of the natural frequencies of the structure, the above procedure was repeated at other symmetrical points (such as point III in column 4 and point III in column 9), and the same conclusion was reached. Therefore, 2.18 Hz is the second natural frequency of the structure with its roof 80 ton lighter than its normal condition, and its correspondent vibration mode is antisymmetrical (torsion). By following the same procedure, natural frequencies of the tower with its roof 80 ton lighter than its normal condition are obtained as shown in Fig. 11.

The second test was carried out after the roof was reconstructed and the structure was in its normal condition. The same procedure as in the first test was followed. The only difference lies in the employment of a 0.196-kN exciter when the signal could not be well obtained for data processing. Fig. 11 shows the first to fourth natural frequencies of the tower in the two roof conditions (with 80-ton difference in roof dead load). It is obvious that the natural frequencies of the structure will decrease when the roof weight increases. Through comparison of the relative amplitudes in the Fourier spectrum of the two sensor points, the structure's vibration configurations corresponding to the first and second modes are also obtained and shown in Fig. 11.

MODEL TEST

The purposes of model tests are (1) to probe the model test method on ancient timber architectures; (2) to investigate the dynamic characteristics in a controllable condition; (3) to verify the conclusions obtained by the on-site measurements; and (4) to obtain the cyclic behavior of timber structures.

Fabrication of Models

Two models were fabricated for the model tests. The partial wooden model (Fig. 12) with the scale of 1:10 was made of *P. koraiensis* by a senior craftsman, trying to keep every detail

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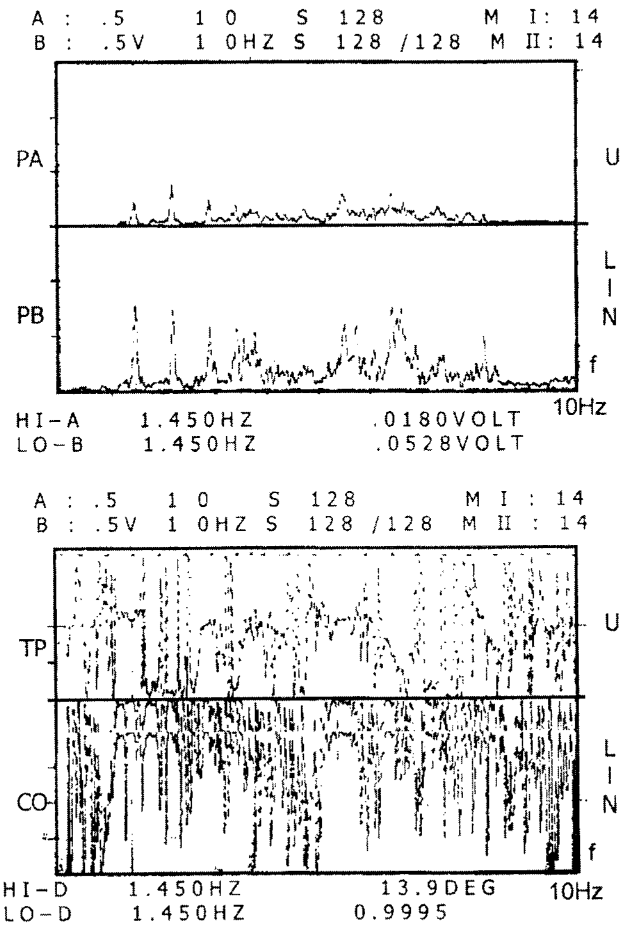


FIG. 10. *P-f*, *TP-f*, and *Co-f* Curves

the same as in the real structure. However, there are two major differences between the model and the real structure: (1) vertical tenon joints were used at the bottom of each column so that the displacement of the model subject to lateral forces could be prevented (Fig. 13); and (2) only a part of the tower was fabricated because of the limitations of funds and lab conditions. This defect limits the wooden model to only representing the vibration mode in the *Y*-direction, not the torsion mode. The integral Perspex, or polymethyl methacrylate (PMMA) model scaled 1:30 (Fig. 14) was made to study the integral characteristics of the tower, but all the connections were fixed by acetone. Similitude ratios of the two models are listed in Table 2. Based on the principle of similitude, all data have been converted to the same basis as the real structure described in the following section.

Multipoint Excitation Test

Vibration Mode and Damping

It is generally difficult to obtain quality signals and pure mode shapes of a complicated structure by single-point excitation. In this paper, multipoint excitation is employed to obtain the true vibration mode of the structure by controlling the excitation frequencies and forces and their phase lag. The key point in this method lies in the excitation force of the two exciters, which need to be equal in frequency and force and have changeable phases. The positions where sensor and exciters were installed are shown in Fig. 9. When the two exciters excite in the same direction, symmetrical modes (in the *Y*-direction) are excited. When they excite in the opposite directions, torsion (antisymmetrical) modes are excited.

By adjusting the excitation frequencies from 0.5 to 20 Hz and processing the data by means of the same method used in the on-site measurement, the natural frequencies and corresponding vibration configurations of the first four modes can be obtained, as shown in Fig. 15. The natural frequencies in the third and fourth modes are much higher than obtained by on-site measurements. The reason will be discussed later in this paper.

The damping ratio can be calculated by a log decrement approach from the free vibration curves of the wooden model and PMMA model. The damping ratio is 0.056 for the wooden model and 0.041 for the PMMA model. Because the connections in the PMMA model are all rigid, it is reasonable that its damping ratio is lower.

Influence of Roof Dead Load

Generally mud and tiles are used in ancient Chinese timber structures for leveling and waterproofing. It makes the roof much heavier than expected. In the on-site test, the influence of roof dead load was obvious. In the model test, influence was obtained by measurement of the structure vibration modes under different roof gravity loads, as shown in Table 3. Apparently, the increase of roof dead loads leads to the decrease of the first and second natural frequencies. However the influence on the third and fourth modes of the wooden model is relatively small because of the insignificant vibration amplitude on the column tops compared with that in the first and second modes (as shown in Fig. 15).

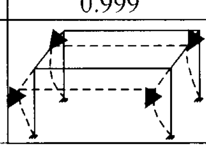
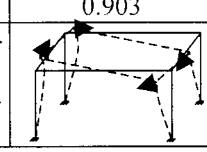
Roof condition (1)	Function (2)	1st mode (vibration in direction Y) (3)	2nd mode (torsion vibration) (4)	3rd mode (vibration in direction Y) (5)	4th mode (torsion vibration) (6)
80 tons lighter in roof	P-f TP-f Co-f	1.450(Hz) 13.9° 0.999	2.175(Hz) -165° 0.999	2.900 (Hz) 4.7° 0.999	3.400(Hz) -165° 0.994
Normal roof condition	P-f TP-f Co-f	1.100(Hz) 14.9° 0.999	1.700(Hz) 164° 0.903	2.725 (Hz) 9.1 0.999	3.100 (Hz) 156° 0.999
Assumed vibration configuration					

FIG. 11. Dynamic Property of Front Tower over North Gate of Xi'an City Wall

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Influence of Connection Rigidity

The integral stiffness of a timber structure increases with the increase of its individual connection rigidity. The structure's natural frequency changes with the change of its integral stiffness. To investigate the influence of connection rigidity, surrounding wooden members were braced so that the stiffness of certain connections could be changed.

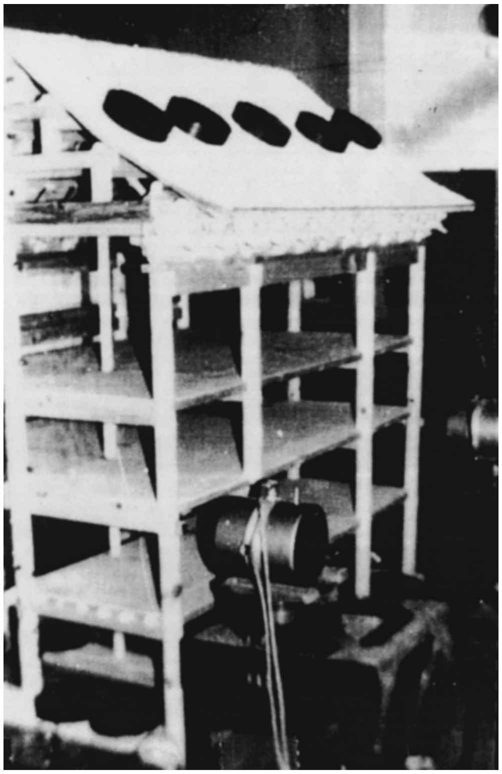

FIG. 12. Partial Wooden Model

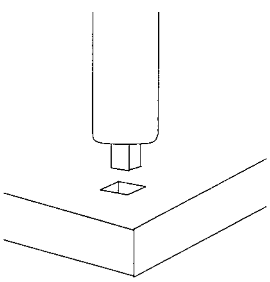

FIG. 13. Tenon Joints Are Used at Bottom of Columns

Special attention was paid to the following three cases: (1) increasing the stiffness of the roof; (2) increasing the stiffness of the dou gong (corbel brackets); and (3) increasing the stiffness of the tenon joints. The natural frequencies of the structure are obtained as shown in Table 4. It is found that there is no significant change to the natural frequency in case 1. Because the roof is stiff enough, further reinforcement cannot change its relative stiffness significantly. Whereas the natural frequencies in cases 2 and 3 were increased by strengthening

TABLE 2. Similitude Ratios of Wooden Model and PMMA Model

Model	Dimension λ_l	Time λ_t	Mass λ_m	Damping ratio λ_h	Force λ_f
Wooden	10	0.1	1,000	1	100
PMMA	30	0.098	10,296.6	1	2,985.6

TABLE 3. Influence of Roof Dead Load on Natural Frequencies of Tower (Hz)

		Assumed Vibration Mode			
Model	Weight (ton)	First mode, vibration in Y-direction	Second mode, torsion vibration	Third mode, vibration in Y-direction	Fourth mode, torsion vibration
Wooden	0	1.46	3.10	8.31	12.68
	80	1.28	2.22	8.67	12.85
	120	1.17	2.04	8.58	12.67
	140	1.11	2.01	8.42	12.66
	160	1.01	1.72	8.39	12.28
	180	0.94	1.71	8.23	12.19
PMMA	0	1.79	2.35	8.02	10.11
	80	1.40	2.07	7.41	9.75
	160	1.20	1.64	6.87	8.98

TABLE 4. Influence of Connection Stiffness on Natural Frequencies of Tower (Hz)

	Assumed Vibration Mode			
Roof stiffness	First mode, vibration in Y-direction	Second mode, torsion vibration	Third mode, vibration in Y-direction	Fourth mode, torsion vibration
Original condition	1.28	2.22	8.67	12.85
Case 1[a]	1.28	2.23	8.70	12.68
Case 2[b]	1.36	2.26	9.66	14.68
Case 3[c]	1.61	2.63	8.84	12.70

[a]Increasing stiffness of roof.
[b]Increasing stiffness of dou gong (corbel brackets).
[c]Increasing stiffness of tenon joints.

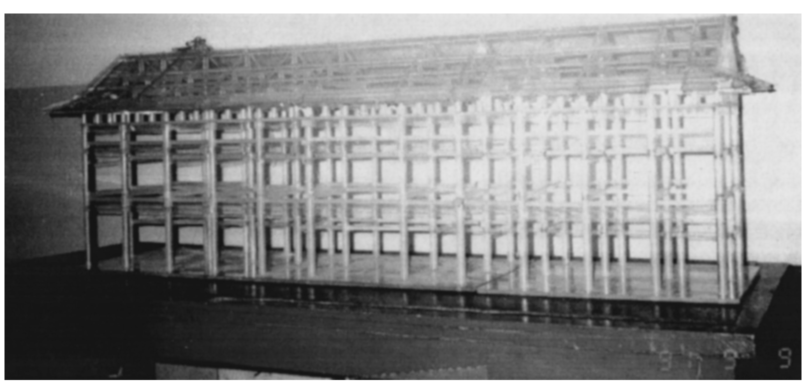

FIG. 14. Integral PMMA Model

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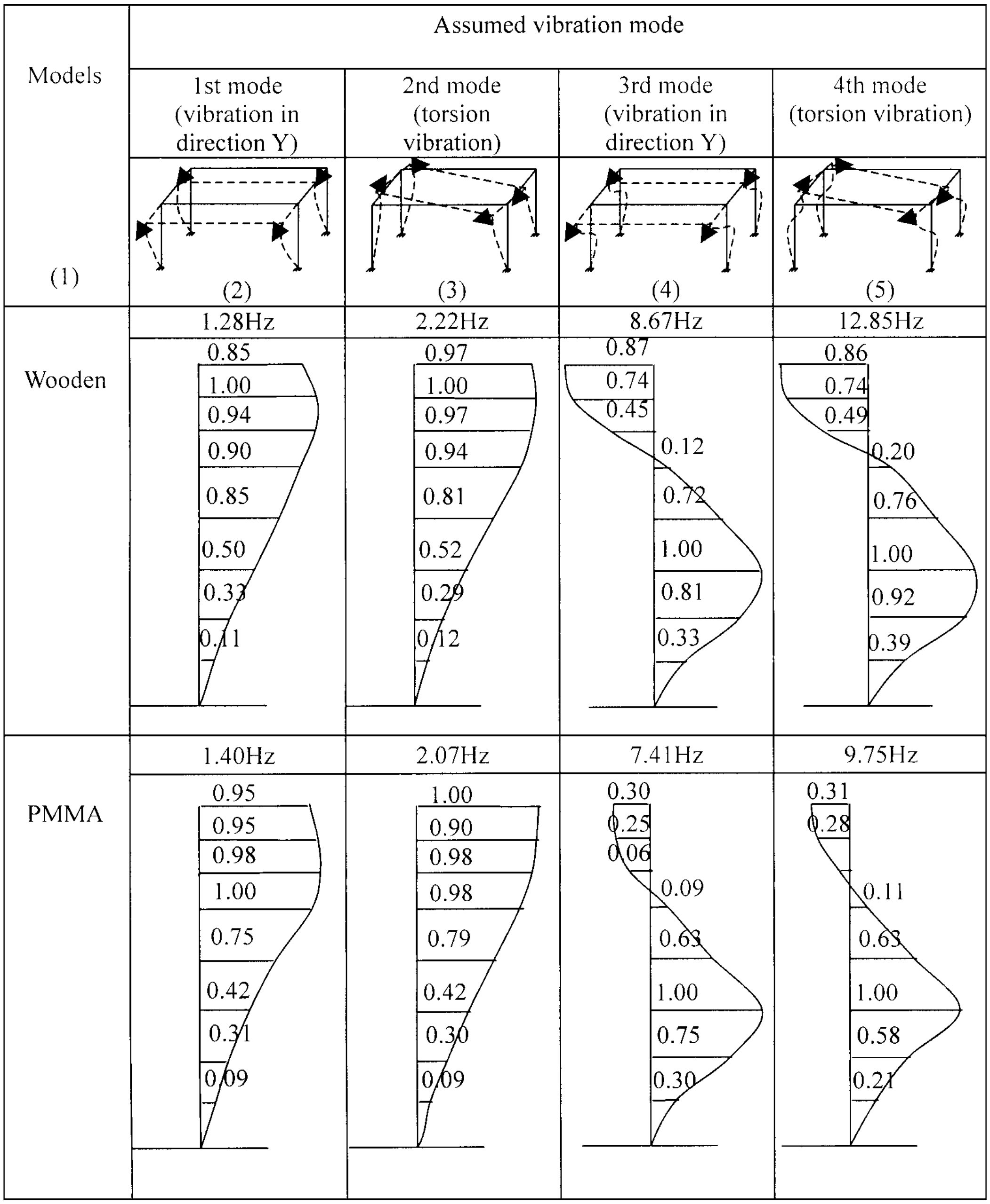

FIG. 15. Vibration Modes Obtained in Model Test

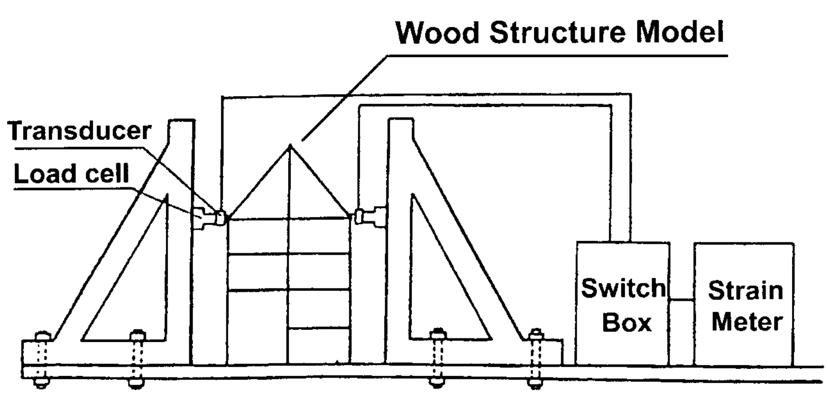

FIG. 16. Test Equipment and Measurement System of Cyclic Loading Test

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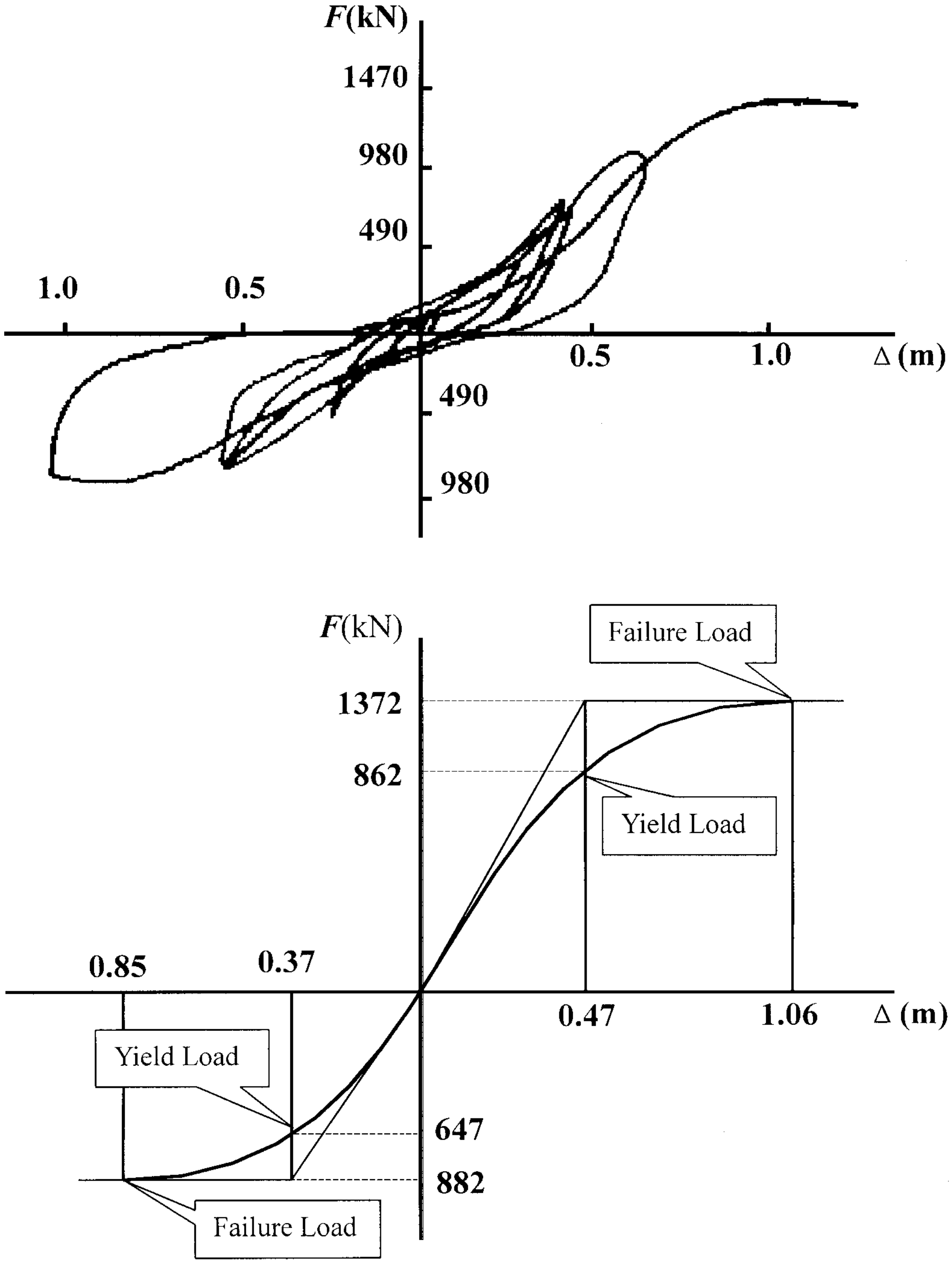

FIG. 17. Hysteresis Loops and Skeleton Curve

TABLE 5. Natural Frequencies and Their Tolerance in On-Site and Model Tests

Method	Roof condition	Assumed Vibration Mode			
		First mode, vibration in Y-direction	Second mode, torsion vibration	Third mode, vibration in Y-direction	Fourth mode, torsion vibration
On-site test	With roof dead load	1.110	1.700	2.725	3.100
	Without roof dead load	1.450	2.175	2.900	3.400
Wooden model	With roof dead load	1.280	2.220	8.670	12.85
	Tolerance	15.3%	30.6%	218.2%	314.5%
	Without roof dead load	1.460	3.100	8.310	12.68
	Tolerance	0.69%	42.5%	186.6%	272.9%
PMMA model	With roof dead load	1.400	2.070	7.410	9.750
	Tolerance	27.3%	21.8%	171.9%	214.5%
	Without roof dead load	1.790	2.350	8.020	10.110
	Tolerance	23.4%	8.0%	8.1%	226.1%

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the dou gong (corbel bracket) and tenon joint. This finding implies that the connections of the dou gong (corbel bracket) and tenon joint are not rigid; they should be regarded as semirigid when the ancient timber structures are studied.

Cyclic Loading Test on Wooden Model

The purposes of the test are to obtain the hysteresis loop and skeleton curve and to observe the structure's performance during loading, especially the responses of dou gong (corbel brackets) and tenon joints.

Fig. 16 shows the loading equipment and statistic measuring system of the wooden model during the cyclic loading test. During the test it was found that the four sensor points on the roof always had similar displacements. This again proved that the roof has very high stiffness. As it was hard to determine the load-carrying capability of the structure, the model was loaded in two directions, with force increased gradually. When the loads in the two directions reached 882 kN (being converted from the model to the real structure), larger relative displacements of the members were observed, the tenon joints were damaged, and the whole structure nearly collapsed. During the test, hysteresis loops and a skeleton curve were obtained as respectively shown in Fig. 17. All the data shown in Fig. 17 were converted from a model test to a true structure. At the beginning of cyclic loading, the stiffness of the structure was not large because the brackets and tenon were moving and adjusting their relative positions. With the increase of loads, the stiffness of the structure increased rapidly because the relative movement of members stopped and the structure became elastic. There is no obvious yield point in the skeleton curve. If the intersection of the tangent line at the origin and the tangent line at the peak load value is considered to be the yield point, the yield load and its corresponding displacement, as well as the failure load and its corresponding displacement, are deduced and shown in the skeleton curve of Fig. 17.

DISCUSSIONS ON VIBRATION MODE OBTAINED FROM ON-SITE AND MODEL TESTS

Natural frequencies obtained from on-site and model tests and their relative tolerance are shown in Table 5. The first natural frequency (Y-direction) of the wooden model is 8% higher than that of the on-site test on average. The main reason is that vertical tenon joints at the bottom of columns of the wooden model, which do not exist in the real structure, enhance the model's stiffness. The second natural frequency (torsion) is on average 36% higher than that of the on-site test, because the wooden model is only a partial model and is short in the X-axial (ridge) direction. Therefore it cannot represent the torsion characteristics of the real structure. The first natural frequency of the PMMA model is 25% (on average) higher than that of the real structure, which results from the rigid connections of the PMMA model. The second natural frequency of the PMMA model is also 15% (on average) higher for the same reason; however, the tolerance is less than that of the wooden model (36%) because the integral model represents the torsion characteristics of the real structure better than the wooden partial model.

The third and fourth frequencies of the model tests were found to be rather different from those of the on-site tests. It is believed that the real third and fourth modes were not identified in the model test. Numerical verification of this conclusion is discussed in Fang et al. (2001).

CONCLUSIONS

- In the full-scale on-site tests, the vibration modes of the front tower over the North Gate of the Xi'an City Wall were measured and analyzed. The first four natural frequencies of the structure are found as 1.10, 1.70, 2.72, and 3.10 Hz, respectively.
- Through multipoint excitation tests on wooden and PMMA models, test methods of ancient timber structures are investigated. These tests validate that the first and second modes obtained by on-site full-scale tests are the natural vibration modes of the tower.
- With reference to the free vibration curve of the wooden model, it is deduced that the damping ratio of the tower is about 0.056. Therefore, the damping ratio of the wooden structure is much larger than that of other structures. This is one reason that they have better seismic performance.
- Heavy roof dead load is one of the unique characteristics of ancient Chinese timber structures. Although increasing the roof dead load may decrease the natural frequency of the structure, it has less influence on those vibration modes with small vibration amplitude at the top of the structure. On the other hand, the increase of the roof dead load magnifies the seismic forces but, at the same time, enhances the friction forces between structural members and connections. Consequently the roof dead load enhances the damping of the structures.
- Roofs of ancient Chinese timber structures have high stiffness. Increasing the stiffness of connections can lead to the increase of the natural frequency of the structure. Therefore, the dou gong (corbel bracket) and tenon joint are semirigid, and have significant influence on the structural stiffness.
- The hysteresis loop and skeleton curve, as well as the working performance of corbel brackets and tenon joints, were acquired through a cyclic loading test. A yield load and initial failure load of the structure are also obtained.
- The method of on-site measurement and modeling suitable to the ancient Chinese timber structures is summarized. Basic information is provided for numerical studies, maintenance, and protection of ancient timber architecture.

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REFERENCES

Brungraber, R. L. (1985). ''Traditional timber joinery: A modern analysis.'' PhD dissertation, Stanford University, Palo Alto, Calif.

Bulleit, W. M., Sandberg, L. B., Drewek, M. W., and O'Bryant, T. L. (1999). ''Behavior and modeling of wood-pegged timber frames.'' *J. Struct. Engrg.*, ASCE, 125(1), 3–9.

Cheng, J. Q. (1985). *Timber*, China Forest Industry Press, Beijing, China, 568–769 (in Chinese).

Chinese ancient architecture. (1985). Tsinghua University Press, Beijing, China (in Chinese).

Chui, Y. H., Ni, C., and Jiang, L. (1998). ''Finite-element model for nailed wood joints under reversed cyclic load.'' *J. Struct. Engrg.*, ASCE, 124(1), 96–103.

Fang, D. P. (1993). ''Experimental and numerical studies on the static and dynamic characteristics of Chinese ancient timber structures.'' Doctoral thesis, Kyushu University, Fukuoka, Japan (in Japanese).

Fang, D. P., Iwasaki, S., Yu, M. H., Shen, Q. P., Miyamoto, Y., and Hikosaka, H. (2001). ''Ancient Chinese timber architecture. II: Dynamic characteristics.'' *J. Struct. Engrg.*, ASCE, 127(11), 1358–1364.

Hayashi, T., et al. (1998). ''Restoring properties of Japanese traditional

J. Struct. Eng., 2001, 127(11): 1348-1357

wooden frame.'' *Proc., World Conf. on Timber Engrg.*, Presses Polytechniques et Universitaires Romandes, Montreux, Switzerland, 750–751.

King, W. S., Yen, J. Y., and Yen, Y. N. (1996). ''Joint characteristics of traditional Chinese wooden frames.'' *Engrg. Struct.*, 18(8), 635–644.

Li, J. (1100). *Yingzao Fashi [Construction method]*, Royal Press, Kaifeng, China (in ancient Chinese).

Liang, S.-C. (1984). *A pictorial history of Chinese architecture*, W. Fairbank, ed., MIT Press, Boston.

Ma, B. J. (1991). *Construction technology of Chinese ancient timber architectures*, Science Press, Beijing (in Chinese).

Maekawa, H., and Kawai, N. (1998). ''Microtremor measurement on Japanese traditional wooden houses which are important cultural properties.'' *Proc., World Conf. on Timber Engrg.*, Presses Polytechniques et Universitaires Romandes, Montreux, Switzerland, 40–47.

Sandberg, L. B., Bulleit, W. M., and Reid, E. H. (2000). ''Strength and stiffness of oak pegs in traditional timber-frame joints.'' *J. Struct. Engrg.*, ASCE, 126(6), 717–723.

Seo, J.-M., Choi, I.-K., and Lee, J.-R. (1999). ''Static and cyclic behavior of wooden frames with tenon joints under lateral load.'' *J. Struct. Engrg.*, ASCE, 125(3), 344–349.

Subcommittee on Wood Research of ASCE Committee on Wood. (1986). ''Structural wood research needs.'' *J. Struct. Engrg.*, ASCE, 112(9), 2155–2165.

Tanikuchi, O., et al. (1991). *Hand book for vibration engineering*, Youkendo Press, Tokyo (in Japanese).

Uchida, A., et al. (1998). ''Dynamic characteristics in Japanese traditional timber buildings.'' *Proc., World Conf. on Timber Engrg.*, Presses Polytechniques et Universitaires Romandes, Montreux, Switzerland, 34–41.

Wang, T. (1992). *Static mechanics of ancient timber structures*, Historical Relic Press, Beijing, China (in Chinese).

Yanagisawa, K., et al. (1998). ''Study on earthquake-protection of traditional Buddhist temples.'' *Proc., World Conf. on Timber Engrg.*, Presses Polytechniques et Universitaires Romandes, Montreux, Switzerland, 572–576.

NOTATION

The following symbols are used in this paper:

E = Young's modulus of elasticity;
γ_{15} = density of wood with 15% moisture content;
λ_h = similitude ratios of damping ratio;
λ_l = similitude ratios of dimension;
λ_m = similitude ratios of mass;
λ_t = similitude ratios of time;
σ_b = bending strength;
σ_c = compression strength;
σ_I = tensile strength; and
τ_b = shear strength.

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